Kisan Alert

Voice & SMS agricultural intelligence for small and marginal farmers

HACK CORE 2026 · Agriculture & Climate-Resilient Farming

Telugu-first · Data-driven · RSK expert-backed

github.com/Siddharth77/smart-kisan

The Problem

- Farmers lose crops to unpredictable monsoons and habit-based decisions—not data.
- Crop choices follow neighbors or tradition, ignoring soil health, groundwater, and rainfall.
- Dry spells arrive without localized irrigation guidance; fertilizer is applied at the wrong time.
- When crops show disease symptoms, there is no fast path from photo to expert help.
- Small and marginal farmers are left out: feature phones, Telugu/Hindi, patchy connectivity.

Who We Serve

- Small & marginal farmers: 1–3 acres, low literacy-friendly UX
- Primary languages: Telugu (demo), Hindi scale path
- Devices: feature phones + occasional smartphone (WhatsApp)
- Demo persona: **Lakshmi Devi**, 2-acre farmer, Hanamkonda, Warangal
- Channels: SMS-style alerts, voice readout, photo upload, RSK callback

Our Solution

- **Three integrated pillars** in one platform:
- 1. **Smart crop recommendation** — soil, rainfall, groundwater → ranked crops + risk scores
- 2. **Real-time advisory** — dry-spell detection → Telugu irrigation & fertilizer alerts
- 3. **Crop health + RSK** — photo diagnosis → auto-ticket to Rythu Seva Kendra when AI is uncertain
- Closed loop: data in → advice out → human expert when needed

Pillar 1 — Crop Recommendation

- Farmer registers village + plot size (acres, season)
- Engine scores crops: soil fit (40%), water need vs availability (35%), season (25%)
- Output: top 3 crops with risk % and reasoning in English + Telugu
- Example: groundnut ranked #1; cotton flagged high water-stress for Hanamkonda kharif
- Data: Warangal soil grid, IMD-style rainfall, groundwater bands

Pillar 2 — Dry-Spell Alerts

- Monitors: days since rain, soil moisture threshold, forecast context
- Trigger: 10+ dry days + moisture < 35% → dry_spell alert
- Farmer inbox (SMS simulator): actionable Telugu message
- Example: “Irrigate lightly in evening; hold fertilizer until rain returns.”
- Voice: Web Speech reads alert aloud; production → Cloud Text-to-Speech

Pillar 3 — Diagnosis & RSK

- Farmer uploads crop photo → disease label + confidence score
- If confidence < 80% → automatic RSK ticket with photo + AI summary
- Expert queue: view, add notes, mark resolved → farmer confirmation
- Demo: leaf_spot (62% confidence) creates ticket; healthy (95%) does not
- Production: Gemini multimodal vision + state RSK MIS webhook

Live Demo Flow

- 1. Register plot → 2. Crop advice → 3. Simulate dry spell
- 4. Photo diagnosis → 5. RSK expert resolves ticket
- Deployed MVP: Next.js on Vercel — zero-config, full E2E working
- Quick demo: Lakshmi Devi pre-seeded; reset anytime via /api/demo/reset
- Dashboard: rainfall stats, soil panel, satellite NDVI map (Warangal)

AI & Technical Approach

- **MVP (now):** Rules engine — reliable, offline-capable, demo-safe
- **Pilot:** Gemini API — crop Q&A, multimodal disease diagnosis
- **Scale:** Vertex AI, Earth Engine (NDVI), BigQuery (IMD + public datasets)
- Architecture: Next.js API routes → pluggable ai-provider interface
- Storage: in-memory (Vercel) / SQLite (local) / Turso or Firestore (prod)

Google Cloud Roadmap

- Gemini API / Vertex AI — recommendations + photo diagnosis
- Firebase — Hosting, Firestore realtime inbox & RSK tickets
- Cloud Functions + Scheduler — automated alert pipeline
- Translation + Text-to-Speech — Telugu/Hindi SMS and voice
- Earth Engine + BigQuery — satellite layers, IMD rainfall, data.gov.in
- Dialogflow — voice/SMS conversational intake for feature phones

Why It Is Deployable

- SMS-first: no app install; works on 2G feature phones
- Serverless: Vercel today → Cloud Run / Firebase tomorrow
- Low cost: rules MVP free; GCP pay-per-use at scale
- RSK-ready: ticket schema maps to Kendra expert workflows
- Progressive AI: rules work without connectivity; Gemini adds depth when available

Pilot & Scale

- **Pilot (6 mo):** Warangal district, 500 farmers, 3 RSK partners
- **Metrics:** alert open rate, crop switch adoption, RSK resolution time
- **District:** 10K farmers, IoT soil sensors, hourly Scheduler alerts
- **State:** Telangana rollout, Dialogflow IVR, ag dept MIS integration
- **National:** 10+ languages, Earth Engine at scale, BigQuery analytics

Thank You

- **Kisan Alert** — data-driven farming in the farmer’s language
- Repo: github.com/Siddharth77/smart-kisan
- Live demo: Vercel deploy (zero-config)
- Team: [Your names here]
- Questions?